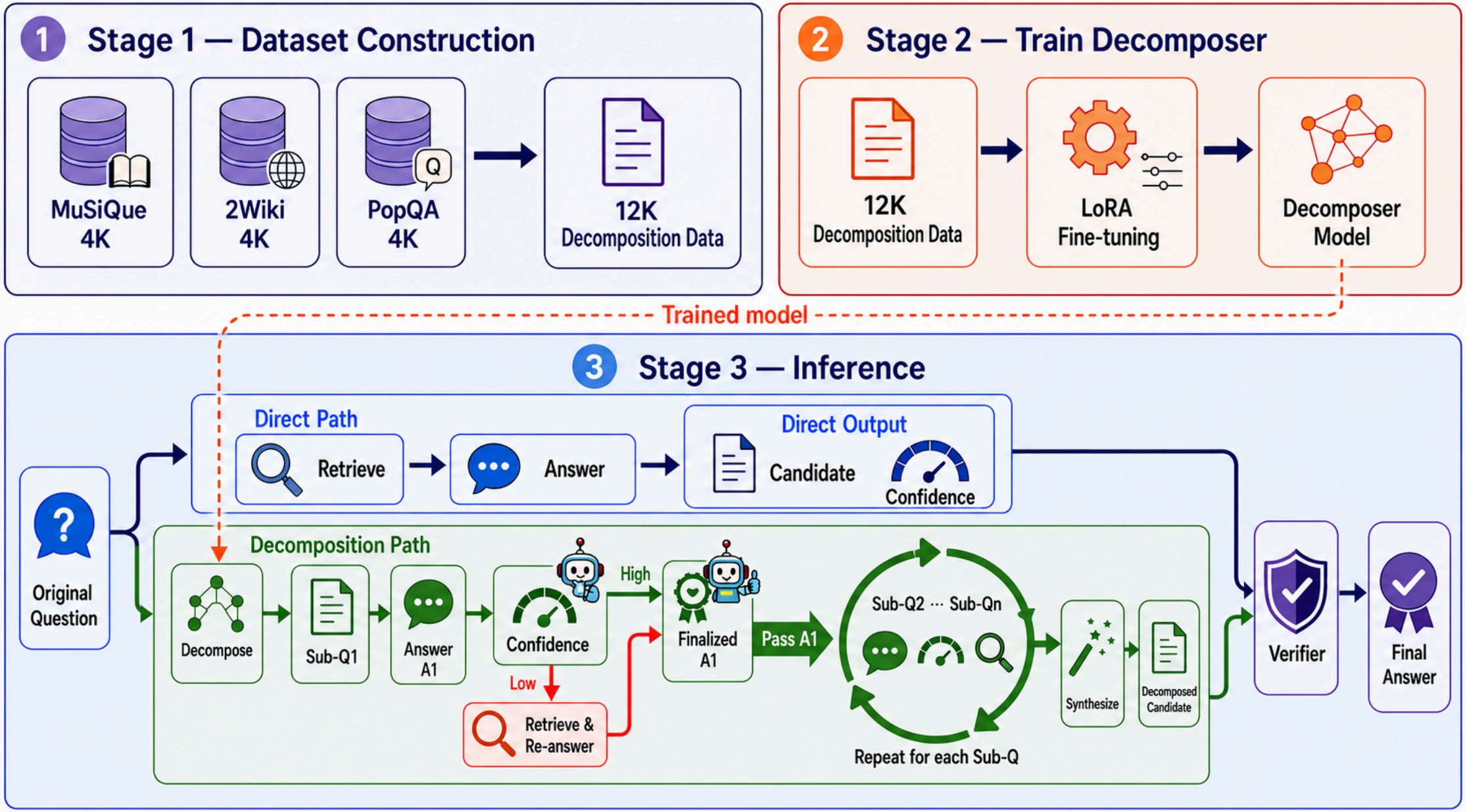

1
Stage 1 — Dataset Construction
MuSiQue 4K
2Wiki 4K
PopQA 4K
Q
12K Decomposition Data
2
Stage 2 — Train Decomposer
12K Decomposition Data
LoRA Fine-tuning
Decomposer Model
Trained model
3
Stage 3 — Inference
Original Question
Direct Path
Retrieve
Answer
Direct Output
Candidate
Confidence
Decomposition Path
Decompose
Sub-Q1
Answer A1
Confidence
High
Low
Retrieve & Re-answer
Finalized A1
Pass A1
Sub-Q2 ··· Sub-Qn
Repeat for each Sub-Q
Synthesize
Decomposed Candidate
Verifier
Final Answer